

Roshan Bhatta

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Education

University of Louisiana Monroe - GPA: 3.0 | Senior

Monroe, LA

Computer Information Systems | Concentration: Data Science

Expected Graduation: Dec 2026

Awards: Ace Scholarship ULM (2022-Present), Emerging Scholar (2022-2023)

Certification: AWS Cloud Practitioner, AWS AI Practitioner, Ethical Hacking Essentials, CCNA Advanced Networking

Relevant Coursework: Information Systems Analysis & Designs, Database Application Development, Information Security Management, Prompt Engineering & Programming with OpenAI, British Airways Data Science Job Simulation

Relevant Technical Skills

Languages: Python, SQL, HTML/CSS, JavaScript

Data Skills: Data wrangling and Visualization, Feature engineering, Statistical Analysis, Hypothesis Testing, Hyperparameter Tuning

Machine Learning: Scikit-learn, Pytorch, Keras, Supervised and Unsupervised Learning, Deep Learning, Computer Vision

Tools: NumPy, Pandas, Matplotlib, Seaborn, Folium, faceTableau, Excel, BeautifulSoup, GridSearchCV, Flask, Git, OpenCV

Relevant Projects

StepSight AI – ACL Injury Risk & Sports Health Analytics Platform

Python, PyTorch, OpenCV, NumPy, Pandas, MRNet / 1-month project / Team of 3

- Built an AI-driven sports injury analytics platform combining **MRNet MRI classification** and **OpenCV-based pose estimation** to assess ACL injury risk.
- Processed **3,000+ MRI images and athlete movement frames**, extracting biomechanical indicators including knee valgus angle and landing symmetry.
- Developed a **Python+PyTorch pipeline** achieving **~87% ACL tear detection accuracy** on validation datasets.

RAG Chatbot

Python, FAISS, Amazon Bedrock, LLMs, Streamlit / Personal Project

- Built a **multi-document RAG chatbot** using FAISS vector search and **AWS Bedrock LLMs** for PDF question answering.
- Implemented **semantic embeddings and document chunking** to enable context-aware retrieval.
- Developed a **Streamlit interface with page-level citations for explainable responses**.

SafeSurf- Privacy Content Blocker

JavaScript, Chrome APIs

- Developed a **Chrome extension** blocking harmful sites using domain and keyword filtering.
- Implemented **local analytics and streak tracking** to reinforce positive browsing habits.
- Designed a **privacy-first architecture** with all processing done locally.

HSRP Multi-VLAN Enterprise Network Design

Cisco Networking, HSRP, VLAN

- Designed a **multi-VLAN enterprise topology with HSRP redundancy** for high availability.
- Implemented **VLAN segmentation and inter-VLAN routing** to improve traffic isolation.
- Simulated failover scenarios achieving **<3-second gateway failover** in Packet Tracer.

Research Papers

Classical vs Quantum Machine Learning (QML) – Comparative Analysis (IEEE Submission Planned)

Python, scikit-learn, QML, Data Analysis

Author | In Progress

- Designed and executed experiments comparing classical ML algorithms with quantum ML techniques on benchmark datasets.
- Evaluated model performance using standard metrics and documented methodological trade-offs between classical and quantum paradigms.

Relevant Leadership and Extracurricular Activities

CAB (Campus Activity Board) | Committee Head

- Grew Assisted with event promotion, logistics, and team coordination.
- Supported planning and execution of campus-wide student events.

GDSC ULM (Google Developer Student Club) | Member

- Led the planning and execution of six **Tech Expo** events, drawing over 200 participants and showcasing cutting-edge technological innovations.